Project Research Notebook



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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Project Research Notebook

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1.0 Purpose and Best Practices:

This document is the official chronological log of all experiments, observations, and manual actions performed within the ACE project testbed. It is the primary source of "ground truth" for what occurred during the research. It is a living document.

Best Practices:

- **Chronological Order:** New entries must always be added to the end of the log.
- **Be Verbose:** Write more than you think you need. Future you will be grateful. Capture thoughts, hunches, and qualitative observations.
- **Timestamp Everything:** At a minimum, capture the date in the format mm/dd/yyyy. If time is relevant, capture the timestamp in UTC to avoid confusion and simplify log correlation across different systems.
- **Record Failures:** Failed experiments are data. Document what went wrong, why you think it went wrong, and what you did to fix it.
- **Link, Don't Duplicate:** Reference external documents (like Test Plans) by their ID, version, and step. Do not paste their contents here.
- **Immutable Raw Data:** Note the location and checksums of raw data. The raw data itself should never be modified. Analysis is performed on copies.

2.0 Experiment Logging

This section will capture each test conducted in your research project testbed. Each execution of your test will be denoted with an “Execution ID”. Use the template provided for the first two entries. Additional entries can be made by using the copy and past from the template at the end of this document. Adjust the template as necessary to support your project. Important: Be detailed as this will help you conduct your analysis, your report, and provide transparency with reproducibility by others.

Note about TC-003:

Test Case 003 will not be included due to time restraints. However, after the internship has concluded, the PI will go back to implement the features needed to execute this test case. This is why TC-003 remains in the PTP-001 document.

2.1 Execution ID: RUN-20251203-001

Date: 2025-12-03
Start Time (UTC): 16:53:47
End Time (UTC): 16:58:47
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-001 – Baseline CAI Idle / No Query Issued

B. Objective:

Establish baseline system performance under idle/no-query conditions as required by TC-001.

C. Summary of Actions:

1. Verified all TC-001 entry criteria per PTP-001 §4.0:
    - All EC2 instances powered on (ece-primary, ece-allocator, log-gen-workstation, cai-node, analyst workstation).
    - Elasticsearch and Kibana containers running on ece-primary (Podman).
    - Filebeat active on log-gen-workstation.
    - PI-Key file present and readable.
    - System resource usage stable (<5% CPU, <10% memory) for 5 minutes prior to measurement.
2. Started system performance collection on ece-primary using:
    - dstat -tcmnd --output /data/raw/RUN-20251203-001_dstat.csv &
3. Allowed the environment to remain completely idle for 5 minutes with no CAI queries, Kibana queries, Elasticsearch Dev Tools activity, or log-generation scripts.
4. Stopped dstat using pkill dstat.
5. Verified presence and integrity of the output file /data/raw/RUN-20251203-001_dstat.csv.

D. Key Observations (Qualitative Data):

CPU: Baseline CPU utilization was very low. Across the run, usr typically stayed in the ~0.7–2.5% range and sys around 0.1–0.6%, with idl generally >97%. There were periodic short spikes where usr reached ~15–24% and sys ~6–9%, causing idl to drop to the 68–78% range for single seconds. These spikes appear consistent with internal ECE background tasks (e.g., coordination, GC) rather than sustained load.

Memory: Memory usage was high but stable. used hovered around 26.9–27.4 GiB, with free between ~0.6–1.0 GiB and cache ~5.4–5.5 GiB. There was no clear upward trend over the sampling window, suggesting a steady-state JVM footprint with Linux using spare RAM for file cache.

Network: Even with no analyst queries, the primary node showed continuous network activity. Aggregate receive/send values were typically in the low hundreds of thousands per sample, with intermittent bursts up to the low millions. This is consistent with routine ECE internal traffic and health checks; no anomalous or runaway network usage was observed.

Disk I/O: Disk read activity was effectively zero throughout the run. Disk write activity was minimal, characterized by occasional small bursts, likely reflecting log writes or periodic housekeeping tasks. No heavy write patterns were observed.

E. Anomalies, Errors, and Deviations from Test Plan:

None observed.

F. Testbed State Changes:

None.

G. Data Pointers:

- **Raw Data Location**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251203-001/

- **Data Integrity Checksums:**
  TC-001_dstat.csv — SHA256:
  965ebf8e3c015d7aef5b083226cf501fa2e607c3c405bc725a1d6341c3769f33

H. Next Steps / Follow-up Actions:

- Proceed to TC-002 to evaluate CAI natural-language query performance for “failed login” detection.
- Use this baseline dataset as a reference point for comparative load analysis in later tests.

2.2 Execution ID: RUN-20251203-002

Date: 2025-12-03
Start Time (UTC): 20:09:41
End Time (UTC): 20:13:22
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-002 – CAI Natural-Language Query: Failed Logins

B. Objective:

Evaluate whether the CAI can correctly translate a natural-language query about failed-login activity into a valid Elasticsearch JSON DSL query and achieve an F1 ≥ 0.95 when retrieving pre-tagged failed-login threat events.

C. Summary of Actions:

1. Verified all entry criteria per PTP-001 §4.0: all EC2 nodes online; Elasticsearch, Kibana, Filebeat operational; PI-Key present; CAI Streamlit UI responsive; and baseline system load stable.
2. Issued the natural-language query into CAI exactly as required by TC-002: “Show me all failed login attempts in the past 24 hours.”
3. Captured the CAI-generated JSON DSL response in its raw, unmodified form and saved it to: /data/raw/RUN-20251203-002/TC-002_cai_output.json
4. Submitted the JSON DSL to Elasticsearch and captured the returned UUID set in: /data/raw/RUN-20251203-002/TC-002_returned_uuids.txt
5. Computed TP, FP, and FN counts, then calculated precision, recall, and F1 score according to definitions in PTP-001 §1.3 and saved it to: /data/raw/RUN-20251203-002/TC-002_f1_scoring.txt

D. Key Observations (Qualitative Data):

- The CAI-generated DSL was syntactically valid and executed successfully on the first attempt.
- No schema errors, missing fields, or invalid operators were observed.
- The JSON was appropriately constrained (no “match_all,” no unintended field expansions), consistent with safe CAI behavior.
- Retrieval behavior closely aligned with PI-Key expectations: the model captured nearly all failed-login events within the 24-hour window.

E. Anomalies, Errors, and Deviations from Test Plan:

- 19 ground-truth failed-login events were not retrieved (FN = 19).
  This does NOT violate the test plan but is just noted.
- No procedural deviations occurred.
- No system errors or CAI fallback responses were encountered.

F. Testbed State Changes:

None.

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251203-002/

- **Data Integrity Checksums:**
  TC-002_cai_output.json — SHA256:
  347f211ded17477a79d0d3fe020b1dba81e74022cac3e0c7194b72defb8bc87d
  TC-002_f1_scoring.txt — SHA256:
  6676203b62c8edb68ea5e5db2222d77a61241d11427c1da4767a07af24196ae7
  TC-002_returned_uuids.txt — SHA256:
  2a471535e081dfb23c281d4b16d6b1c96d79c220640cd5b8341bf8f72e48bbd5

H. Next Steps / Follow-up Actions:

- Proceed to TC-003 (Suspicious IP Lookup), using the same data-capture and evaluation workflow.
- Investigate the 19 false negatives to determine what happened.
- Investigate how to make the generated logs more complicated as to make this simple task more challenging, while also improving the CAI’s query skills to keep up with the added complexity.

2.4 Execution ID: RUN-20251203-004

Date: 2025-12-03
Start Time (UTC): 21:46:32
End Time (UTC): 21:50:35
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-004 – Lateral Movement Chain Detection

B. Objective:

Evaluate whether the CAI can correctly generate an Elasticsearch JSON DSL query capable of detecting synthetic multi-step lateral movement chains (SSH, SMB, RDP, PsExec, WMI, etc.) and whether it meets the required accuracy threshold of F1 ≥ 0.90.

C. Summary of Actions:

1. Verified entry criteria per PTP-001 §4.0: all EC2 nodes running; Elasticsearch, Kibana, Filebeat operational; CAI responsive; PI-Key present; system load stable.
2. Entered the natural-language query into the CAI:
   "Show me lateral movement between hosts."
3. Captured the CAI-generated JSON DSL and saved it to:
   /data/raw/RUN-20251203-004/TC-004_cai_output.json
4. Submitted the JSON DSL to Elasticsearch and saved the raw output to:
   /data/raw/RUN-20251203-004/TC-004_es_response.json
5. Extracted UUIDs returned by Elasticsearch and saved to:
   /data/raw/RUN-20251203-004/TC-004_returned_uuids.txt
6. Extracted ground-truth lateral movement UUIDs from PI-Key and saved to:
   /data/raw/RUN-20251203-004/TC-004_ground_truth_uuids.txt
7. Computed TP, FP, FN, and calculated Precision, Recall, and F1 score.

D. Key Observations (Qualitative Data):

- CAI successfully produced a syntactically valid JSON DSL query with no schema violations.
- The model over-generalized, using very broad filters that pulled in thousands of unrelated authentication or network events.
- Lateral movement is a multi-stage behavior requiring:

    - Correlation across hostnames
    - Sequence-aware event logic
    - Multiple event categories (authentication, network, process, RDP/SSH)
- The CAI lacks sufficient structural understanding to build proper multi-field, multi-step correlation filters on its own.
- Despite high false positive volume, the model retrieved all real lateral movement events (FN = 0), indicating over-selection rather than under-selection.

E. Anomalies, Errors, and Deviations from Test Plan:

The CAI query produced an extremely wide match condition, leading to 5,068 false positives.
No JSON syntax errors or execution errors occurred.
No deviations from procedural steps.

F. Testbed State Changes:

None

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251203-004/

- **Data Integrity Checksums:**
  TC-004_cai_output.json — SHA256:
  3aa3474f223541649d65c57fa850bc0c45a1ddae0c6444e75aed1f3bd040e92b
  TC-004_f1_scoring.txt — SHA256:
  baed25fea3d5ed107ee7a5aaa48cc30536f452231ae14e61b8ff20d78db2b5ac
  TC-004_returned_uuids.txt — SHA256:
  dee50b7a23237fa29c208d0f27d0b1f7b6b41a60d66531e5a5f972d18db74f51

H. Next Steps / Follow-up Actions:

- Investigate root causes of over-selection:
  Evaluate whether CAI used fields like event.category: authentication too broadly instead of specific markers such as event.action: rdp_login, process.executable: psexec.exe, or suspicious cross-host sequences.
- Enhance the CAI prompt template:
  Add structural rules that force the model to include:

    - Specific remote access protocols (RDP, SSH, SMB)
    - Sequence/causality requirements
    - Host-to-host linkage fields (source.ip, host.name, destination.ip)
- Add schema awareness to CAI:
  Consider injecting ECS field constraints and known lateral movement logic directly into the model's system prompt.
- Re-run TC-004 after improvements.
- Document failure:
  F1 = 0.5858, which is below the success threshold (0.90); therefore TC-004 fails as currently implemented.

2.5 Execution ID: RUN-20251206-005

Date: 2025-12-06
Start Time (UTC): 18:25:32
End Time (UTC): 18:26:12
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-005 – Invalid / Ambiguous Query Handling

B. Objective:

Ensure CAI safely handles malformed analyst inputs.

C. Summary of Actions:

1. Verified entry criteria per PTP-001 §4.0: all EC2 nodes running; Elasticsearch, Kibana, Filebeat operational; CAI responsive; PI-Key present; system load stable.
2. Entered the natural-language query into the CAI:
   "Find the suspicious things but not too many."
3. Captured the CAI-generated response and saved it to: /data/raw/RUN-20251206-005/TC-005_cai_output.txt

D. Key Observations (Qualitative Data):

The CAI performed flawlessly. No malformed query was sent to Elasticsearch, and it requested clarification instead of hallucinating information to make the user happy.

E. Anomalies, Errors, and Deviations from Test Plan:

None.

F. Testbed State Changes:
No services were restarted or altered during the test.

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251206-005/

- **Data Integrity Checksums:**
  TC-005_cai_output.txt — SHA256:
  02d789b1da2070cb953cb492cfe8937e0a882d416bcba0917896863894914 62f

H. Next Steps / Follow-up Actions:

- Proceed to TC-006.

2.6 Execution ID: RUN-20251207-006

Date: 2025-12-07
Start Time (UTC): 01:10:42
End Time (UTC): 01:19:12
Researcher(s): Koi McFarland

A. Linked Documents:
Test Plan: PTP-001 (v1.3)
Test Case(s): TC-006 – Query Robustness – Synonym & Paraphrase Variation Test

B. Objective:
Evaluates the linguistic robustness of the CAI by giving minor variations in natural-language phrasing to determine if it degrades the detection accuracy.

C. Summary of Actions:

1. Verified entry criteria per PTP-001 §4.0: all EC2 nodes running; Elasticsearch, Kibana, Filebeat operational; CAI responsive; PI-Key present; system load stable.
2. Issue the following semantically equivalent queries into the CAI interface:
   - “Show me all unsuccessful login attempts from the past hour.”
   - “List authentication failures for user accounts within the past hour.”

- "Which events had login problems in the last hour?"
- "Display events in the last hour where someone tried to log in but couldn't."
- "Failed authentication logs from the past hour."

3. For each query:
    - Record the CAI-generated JSON DSL.
    - Submit the DSL to Elasticsearch.
    - Extract returned event UUIDs.
    - Compare with PI-Key ground truth labeled failed-login events.
    - Compute precision, recall, and F1 score for each phrasing.
    - Refresh the CAI page before completing the next prompt.

D. Key Observations (Qualitative Data):

The CAI correctly interpreted each query in the same exact way.

E. Anomalies, Errors, and Deviations from Test Plan:

None.

F. Testbed State Changes:

The prompts were each changed to constrain the time-frame to an hour before the prompt was submitted. This is because the log generation code had been modified earlier in the day, and the PI wanted to ensure the CAI was only accessing new and updated logs.

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251207-005/

- **Data Integrity Checksums:**
  TC-006_cai_output_1.json — SHA256:
  8f94b3840ad930d4b37e7536600dbd4ca7dec6ab1f3bb0dd908ad50c1fddbd43
  TC-006_cai_output_2.json — SHA256:
  8f94b3840ad930d4b37e7536600dbd4ca7dec6ab1f3bb0dd908ad50c1fddbd43
  TC-006_cai_output_3.json — SHA256:
  8f94b3840ad930d4b37e7536600dbd4ca7dec6ab1f3bb0dd908ad50c1fddbd43
  TC-006_cai_output_4.json — SHA256:
  8f94b3840ad930d4b37e7536600dbd4ca7dec6ab1f3bb0dd908ad50c1fddbd43
  TC-006_cai_output_5.json — SHA256:

8f94b3840ad930d4b37e7536600dbd4ca7dec6ab1f3bb0dd908ad50c1fddbd43
TC-006_f1_scoring_1.txt — SHA256:
0a25da4aaec70b850d8fb23d381800566a1cc61f15343ea08f7d19664324e883
TC-006_f1_scoring_2.txt — SHA256:
64c0410d8707fac104b201ad7588eff9138d4e084afa19267d8f975561b77ae8
TC-006_f1_scoring_3.txt — SHA256:
76811478e9e5e65ef6e7a95c39cbfa3c6a02e6074affadc4c4898cd029a94a9a
TC-006_f1_scoring_4.txt — SHA256:
ddcb55456b867030750d753ea5b9f86703b9082e89492d9d5041214f7fbe4105
TC-006_f1_scoring_5.txt — SHA256:
794dcccfa17ba4de7af60e371bff42cac008b59b0173b0c8f30949f3c7815063
TC-006_returned_uuids_1.txt — SHA256:
631ccbceb37562875b26674338023042354da3ded89d4abb825c0d26c018268c
TC-006_returned_uuids_2.txt — SHA256:
2281c957ed5d31bf07ae1fc515321972b994948cc3ee75308305df862b31b82b
TC-006_returned_uuids_3.txt — SHA256:
776f22c7f72920d94aadde0e6cc977ae750a21246f4affcacda98499f6d4b330
TC-006_returned_uuids_4.txt — SHA256:
bfffa6174f4a8bd88a496356bb29c3952cd696e685f8d1253483ae062a24c2ac
TC-006_returned_uuids_5.txt — SHA256:
c1a5de77063fca1fe92d1bd7a1e022f538fa8e5ed77adc250f4418a8e422f31e

H. Next Steps / Follow-up Actions:

- None

2.7 Execution ID: RUN-20251208-002

Date: 2025-12-08
Start Time (UTC): 01:00:12
End Time (UTC): 01:05:22
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-002 – CAI Natural-Language Query: Failed Logins

B. Objective:

Evaluate whether the CAI can correctly translate a natural-language query about failed-login activity into a valid Elasticsearch JSON DSL query and achieve an F1 ≥ 0.95 when retrieving pre-tagged failed-login threat events.

C. Summary of Actions:

1. Verified all entry criteria per PTP-001 §4.0: all EC2 nodes online; Elasticsearch, Kibana, Filebeat operational; PI-Key present; CAI Streamlit UI responsive; and baseline system load stable.
2. Issued the natural-language query into CAI exactly as required by TC-002: "Show me all foreign failed login attempts in the past 2 hours."
3. Captured the CAI-generated JSON DSL response in its raw, unmodified form and saved it to: /data/raw/RUN-20251208-002/TC-002_cai_output.json
4. Submitted the JSON DSL to Elasticsearch and captured the returned UUID set in: /data/raw/RUN-20251208-002/TC-002_returned_uuids.txt
5. Computed TP, FP, and FN counts, then calculated precision, recall, and F1 score according to definitions in PTP-001 §1.3 and saved it to: /data/raw/RUN-20251208-002/TC-002_f1_scoring.txt

D. Key Observations (Qualitative Data):

- The CAI ultimately generated a structurally valid and semantically appropriate query that constrained results by:
    - Authentication failure indicators
    - Time window (2 hours)
    - Foreign IP geolocation fields
- Once corrected, retrieval behavior aligned closely with PI-Key expectations, indicating correct conceptual understanding of the task.
- The model demonstrated sensitivity to ECS keyword-field requirements, which directly impacted query executability.

E. Anomalies, Errors, and Deviations from Test Plan:

- **Initial Execution Failures (Non-Final Runs):**
    - **Failure 1:** The CAI used `country_iso_code` without the `.keyword` suffix, resulting in a field-mapping mismatch.
    - **Failure 2:** The CAI incorrectly structured the `range` clause outside of `bool.filter`, violating explicit prompt constraints.

- These failures occurred **before** the final successful execution and were corrected through re-issuance of the same query without modifying the dataset.
- **Final Metrics (Successful Execution):**
    - False Negatives (FN): 33
    - False Positives (FP): 1
    - These values do not violate TC-002 success criteria but are documented for trend analysis.
- No procedural deviations from TC-002 occurred.

F. Testbed State Changes:

None.

G. Data Pointers:

Raw Data Location:

All raw artifacts for this run are stored under:
/data/raw/RUN-20251208-002/

- **Data Integrity Checksums:**
  TC-002_cai_output.json — SHA256:
  702d16873e20102235944139c3c2cdc92e7e12aef81217cce67643db6ea54a12
  TC-002_f1_scoring.txt — SHA256:
  df671d02f3bde74116a4f85aeca04d117fe1f064722034d0e8d335a800ec094d
  TC-002_returned_uuids.txt — SHA256:
  7e4e045b42d29dfbf4be38f466adce8617e905feaa4531efca8d1d01e975cc19

H. Next Steps / Follow-up Actions:

- Investigate whether ECS keyword-field constraints can be enforced more rigidly in the CAI prompt template.
- Track frequency of `range`-placement violations across subsequent runs.

2.8 Execution ID: RUN-20251208-003

Date: 2025-12-08
Start Time (UTC): 01:50:34
End Time (UTC): 02:22:23
Researcher(s): Koi McFarland

A. Linked Documents:

Test Plan: PTP-001 (v1.3)
Test Case(s): TC-003 – Suspicious IP Lookup

B. Objective:

Test CAI's ability to generate correct DSL for correlating events involving foreign IPs.

C. Summary of Actions:

1. Verified all entry criteria per PTP-001 §4.0: all EC2 nodes online; Elasticsearch, Kibana, Filebeat operational; PI-Key present; CAI Streamlit UI responsive; and baseline system load stable.
2. Issued the natural-language query into CAI exactly as required by TC-003: "Find events from foreign IP addresses within the past hour."
3. Captured the CAI-generated JSON DSL response in its raw, unmodified form and saved it to: /data/raw/RUN-20251208-003/TC-003_cai_output.json
4. Submitted the JSON DSL to Elasticsearch and captured the returned UUID set in: /data/raw/RUN-20251208-003/TC-003_returned_uuids.txt
5. Computed TP, FP, and FN counts, then calculated precision, recall, and F1 score according to definitions in PTP-001 §1.3 and saved it to: /data/raw/RUN-20251208-003/TC-003_f1_scoring.txt

D. Key Observations (Qualitative Data):

- The CAI correctly identified geolocation-based filtering logic and constrained results to foreign IP ranges.
- Retrieval behavior demonstrated high recall, indicating that the CAI did not under-select relevant events.
- False positives primarily resulted from logs containing valid foreign IP fields but lacking threat-relevant context.

E. Anomalies, Errors, and Deviations from Test Plan:

- 5 failed-login events were not retrieved (FN = 5).
  There was 11 false-positives (FP = 11)
  This does NOT violate the test plan but is just noted.
- There were no deviations from the test plan.

F. Testbed State Changes:

The test-bed was changed to modify the CAI prompt to include an extra instruction to filter out IPs that do not exist when it comes to IP related queries. If there is no IP, then the log should not be included in the results.

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251208-003/

- **Data Integrity Checksums:**
  TC-003_cai_output.json — SHA256:
  28f73e34fa428e27f753cfe8bb049e4716191fc6d1874c1554f805d0e3ba5961
  TC-003_f1_scoring.txt — SHA256:
  5347da61126a66f68434c0aa512d0534baf2b3aaae75602e40a5519384a1996d
  TC-003_returned_uuids.txt — SHA256:
  2e3297a220c7a41c56fc45fee0d0f9ef01339834294de8f6cab05648f0299f77

H. Next Steps / Follow-up Actions:

- Continue to TC-004.

2.9 Execution ID: RUN-20251208-004

Date: 2025-12-08
Start Time (UTC): 01:50:34
End Time (UTC): 02:22:23
Researcher(s): Koi McFarland

A. Linked Documents:
Test Plan: PTP-001 (v1.3)
Test Case(s): TC-004 – Lateral Movement Chain Detection

B. Objective:
Validate CAI's ability to detect multi-step RDP/SSH lateral movements.

C. Summary of Actions:

1. Verified all entry criteria per PTP-001 §4.0: all EC2 nodes online; Elasticsearch, Kibana, Filebeat operational; PI-Key present; CAI Streamlit UI responsive; and baseline system load stable.

2. Issued the natural-language query into CAI exactly as required by TC-002:
   “Find multi-stage lateral movements within the past hour that come from foreign IPs.”
3. Captured the CAI-generated JSON DSL response in its raw, unmodified form and saved it to:
   /data/raw/RUN-20251208-004/TC-004_cai_output.json
4. Submitted the JSON DSL to Elasticsearch and captured the returned UUID set in:
   /data/raw/RUN-20251208-004/TC-004_returned_uuids.txt
5. Computed TP, FP, and FN counts, then calculated precision, recall, and F1 score according to definitions in PTP-001 §1.3 and saved it to:
   /data/raw/RUN-20251208-004/TC-004_f1_scoring.txt

D. Key Observations (Qualitative Data):

- The CAI successfully generated a syntactically valid query incorporating:
  - Temporal constraints
  - Foreign IP filtering
  - Lateral movement–related event categories
- The model required prompt phrasing closely aligned with examples in the system prompt, showing limited semantic generalization for lateral movement concepts.
- Once properly constrained, retrieval behavior gave a good F1 score.

E. Anomalies, Errors, and Deviations from Test Plan:

- 9 failed-login events were not retrieved (FN = 9).
  There was 8 false-positives (FP = 8)
  This does NOT violate the test plan but is just noted.
- The prompt had to be adjusted to get the Model to understand what was being requested. It did not recognize “lateral movements between hosts” instead it needed me to stick more closely to the examples given in the model prompt.

F. Testbed State Changes:

None

G. Data Pointers:

- **Raw Data Location:**
  All raw artifacts for this run are stored under:
  /data/raw/RUN-20251208-004/

- **Data Integrity Checksums:**
  TC-004_cai_output.json — SHA256:
  2cdc1c1b76e894e899a3df8354fdf838e18c77422a38ad6c0f6712a3be9b763f
  TC-004_f1_scoring.txt — SHA256:
  c243191910ccf41f548c570f231d5120a62dc24286571f101e287bf645dd8f46
  TC-004_returned_uuids.txt — SHA256:
  d8850b79df52a715858f7befb8a1354a66dce22e7b2ab13e9af109a2bf73507a

H. Next Steps / Follow-up Actions:

- Analyze prompt dependency effects on lateral movement detection.
- Compare TC-004 performance before and after foreign-IP constraints to quantify precision gains.

3.0 References

ID #	Document Title	Location / Link
PTP-001	Projest Test Plan	./docs/docx/Projext-Test-Plan.docx

3.0 Change Log

Ver.	Date	Description of Change	Author
v1.1	2025-12-03	Filled out document (first time)	Koi McFarland
v1.2	2025-12-06	Completed TC 5 and 6. Added comments on TC 3.	Koi McFarland
v1.3	2025-12-08	Redid TC 2, 3, 4.	Koi McFarland

4.0 Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer	PI	2025-12-17	Final Deliverable
Faculty Advisor			
DoD/ACE Mentor			